Chapter 5

Results

This chapter presents the experimental acquisition, model-comparison and decoding results. No simulation study was performed; consequently, a separate simulation-results section is not included. Development results are distinguished from the reserved MAIN08–MAIN09 evaluation. Text scores measure recovery of the intended targets because reference Morse labels were assigned by pulse-count alignment rather than independent verification of the transmitted symbols.

5.1 Acquisition integrity and primary dataset

The main collection contained 675 recording attempts, including continuation captures associated with MAIN01B. Of these, 674 were nonempty and one was empty. Across the nonempty attempts, 5,390,612 samples were recorded, corresponding to approximately 109.06 minutes of signal. The timestamp-derived effective acquisition rate ranged from 823.540 to 823.993 samples/s, with a median of 823.694 samples/s. These measurements describe the acquired data, rather than the nominal 833 Hz device setting or the subsequent 833 Hz analysis grid.

The acquisition audit found no missing or duplicate sample indices, nonmonotonic timestamps, key/motor-command discrepancies or reported logger CRC errors in the main recordings. No near-full-scale samples were identified by the absolute raw-count threshold of 32760. The recorded status flags did not indicate acquisition overruns, I^2C errors or queue overflow. These checks support digital recording integrity; they do not establish an alias-free channel or manufacturer-compliant actuator drive.

After the complete inclusion policy was applied, 638 captures formed the primary dataset and 37 attempts were excluded from that cohort. Exclusions included extra or superseded captures and unresolved metadata, rather than exclusively acquisition failures. Ten planned target/condition slots remained omitted. The retained primary captures contained 5,058,055 sensor samples and 51 complete clean/table-tap pairs. Table 5.1 summarises the fixed partitions. All groups came from P01 and the same local collection day.

Table 5.1: Primary dataset used for model development and evaluation. Character counts include spaces in the intended targets.

Partition	Session groups	Captures	Intended characters
Training	MAIN01–MAIN06	423	1191
Validation	MAIN07	71	204
Reserved test	MAIN08–MAIN09	144	410
Total	MAIN01–MAIN09	638	1805

5.2 Development comparisons

5.2.1 Original event features

Table 5.2 compares the duration baseline and three ML methods using the original Z-envelope event features. Grouped cross-validation (CV) pooled the predictions obtained by holding out each of the six training sessions in turn. It therefore assessed 423 held-out recordings, rather than reporting fitting-set performance. The MAIN07 column reports a separate development comparison after fitting on all six training sessions.

Table 5.2: Development performance with the original event features. Lower CER is better. Exact values are counts of correctly recovered complete intended messages.

	Grouped training CV		MAIN07 validation	
Method	CER (%)	Exact	CER (%)	Exact
Duration baseline	15.62	328/423	11.76	59/71
Logistic regression	13.35	339/423	8.33	61/71
Original RBF SVM	13.27	339/423	8.33	62/71
Random forest	12.76	341/423	11.76	60/71

Random forest had the lowest pooled original-feature CV CER, 12.76%, compared with 15.62% for the duration baseline. On MAIN07, logistic regression and SVM each produced 17 character edits, corresponding to 8.33% CER; SVM recovered one more complete message. The model ranking was therefore not identical across the two development evaluations.

The development tables report the archived development computations. The final-test tables below use the locally reproduced final evaluation. Their software environments differed, and the small observed random-forest reproduction difference is reported with the final results rather than silently combining the two final runs.

5.2.2 Channel-feature experiment

The channel experiment compared the original envelope features, additional Z-waveform statistics, XYZ acceleration statistics and all six acceleration/gyroscope channels. Table 5.3 gives the full comparison. All configurations shared Z-based detection and timing normalisation; only classifier features changed.

The enriched Z-only SVM had the lowest pooled CV CER in this experiment, 12.17%, with 342/423 exact messages. Its MAIN07 CER was 7.84%, with 62/71 exact messages. XYZ SVM achieved a lower MAIN07 CER of 7.35%, but a higher CV CER of 13.43%. Adding gyroscope features increased CER relative to XYZ acceleration for all three classifiers in both evaluations. These observations are specific to the tested band, features and shared detector.

The enriched Z-only SVM was selected as the primary final candidate using the development evidence. The duration baseline and the three original-feature classifiers were retained as declared comparators before the reserved sessions were evaluated.

Table 5.3: Channel-feature comparison using fixed classifier settings. Each CER is pooled across the recordings in its evaluation partition. This experiment does not compare alternative-axis detectors.

Classifier	Feature configuration	CV CER (%)	MAIN07 CER (%)
Logistic regression	Original envelope	13.35	8.33
Logistic regression	Enriched Z	13.10	7.84
Logistic regression	XYZ acceleration	12.93	9.80
Logistic regression	XYZ + gyroscope	13.69	11.76
RBF SVM	Original envelope	13.27	8.33
RBF SVM	Enriched Z	12.17	7.84
RBF SVM	XYZ acceleration	13.43	7.35
RBF SVM	XYZ + gyroscope	14.61	8.82
Random forest	Original envelope	12.76	11.76
Random forest	Enriched Z	13.01	11.76
Random forest	XYZ acceleration	13.18	12.25
Random forest	XYZ + gyroscope	13.85	13.24

5.3 Reserved-session decoding performance

The final evaluation used 144 recordings from MAIN08–MAIN09 and 410 intended reference characters. All five methods remained fitted on MAIN01–MAIN06. Table 5.4 reports the fixed comparison, while Figure 5.1 displays CER graphically.

Table 5.4: Final results from the verified local run. The enriched Z SVM was the preselected primary candidate; the original SVM was a declared comparator. CER includes spaces and invalid-token question marks.

Method	Edits	CER (%)	Exact messages	Exact (%)
Duration baseline	55	13.41	119/144	82.64
Logistic regression	41	10.00	120/144	83.33
Original RBF SVM	37	9.02	122/144	84.72
Random forest	56	13.66	120/144	83.33
Enriched Z SVM (primary)	47	11.46	118/144	81.94

The original SVM achieved the lowest observed test CER, 9.02%, and recovered 122/144 complete intended messages (84.72%). It produced 18 fewer edits than the duration baseline, a descriptive reduction of 32.7% in edit count. The preselected enriched SVM produced 47 edits (11.46% CER) and recovered 118/144 messages. Its development-set advantage did not persist on the reserved sessions. This outcome did not change the predeclared primary selection.

The local random forest had 56 edits and 13.66% CER. The archived reference evaluation had 55 edits and 13.41% CER; two predicted messages differed, while its exact-message count remained 120/144. The source-code hashes matched, but the fitted artifacts and software versions differed. All other methods' final summary scores matched the reference run. The local values are used consistently in this chapter. No statistical superiority claim is attached to the observed differences.

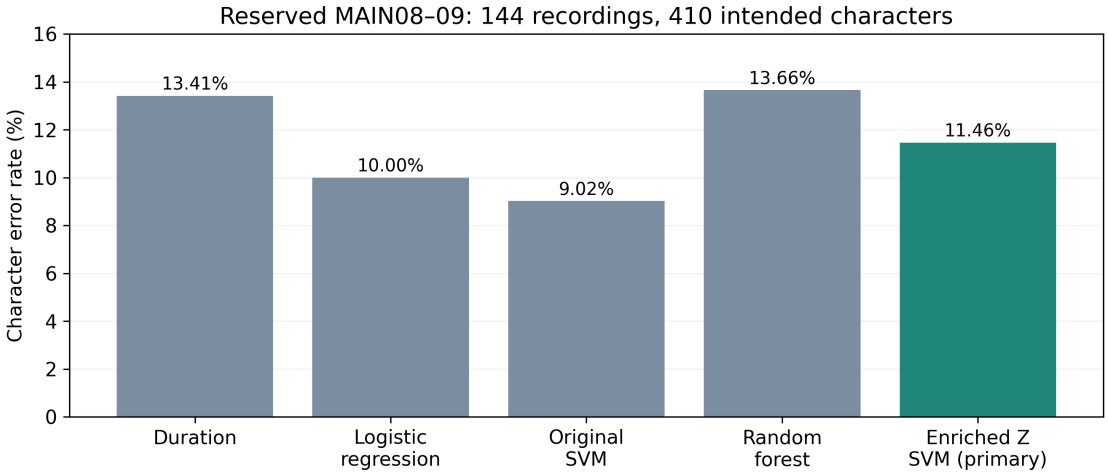


Figure 5.1: Character error rate on the reserved sessions. The highlighted bar identifies the primary candidate, rather than the lowest observed test score. All methods used the same 144 captures.

5.4 Event detection and classification confusion

5.4.1 Detection coverage

The shared detector matched 1308 of 1310 key-reference pulses, with two unmatched reference pulses and no extra detections under the 150 ms one-to-one onset-matching rule. Matching precision was 100%, recall was 99.85% and F1 was 99.92%. Because detection was shared, these values were identical for all five final methods. They describe association with key timing, not independently measured physical vibration boundaries.

5.4.2 Dot and dash classification

Figure 5.2 shows the confusion matrix for the 1308 matched pulses. Rows denote intended classes and columns predicted classes. Each cell gives its count and the percentage of its reference row. The original and enriched SVMs had the same aggregate pulse confusion matrix, although this does not imply identical predictions at every pulse position.

Of 696 matched intended dots, 690 were predicted as dots and six as dashes. Of 612 matched intended dashes, 602 were predicted as dashes and ten as dots. Overall matched-pulse agreement was 1292/1308 (98.78%). This conditional event-label agreement is distinct from the 84.72% and 81.94% complete-message recovery of the two SVMs.

5.4.3 Element, character and word gaps

Gap labels were compared only where adjacent detections matched consecutive reference pulses. This produced 1163 comparable gaps per method. Figure 5.3 compares the two SVM gap heads using the same orientation and percentage convention as the pulse matrix.

The original SVM made 15 comparable-gap disagreements: six intended element gaps were predicted as character gaps, eight character gaps as element gaps and one word gap as a character gap. Its overall comparable-gap agreement was 1148/1163 (98.71%). The enriched SVM made 20 disagreements, giving 1143/1163 agreement (98.28%). The additional disagreements comprised

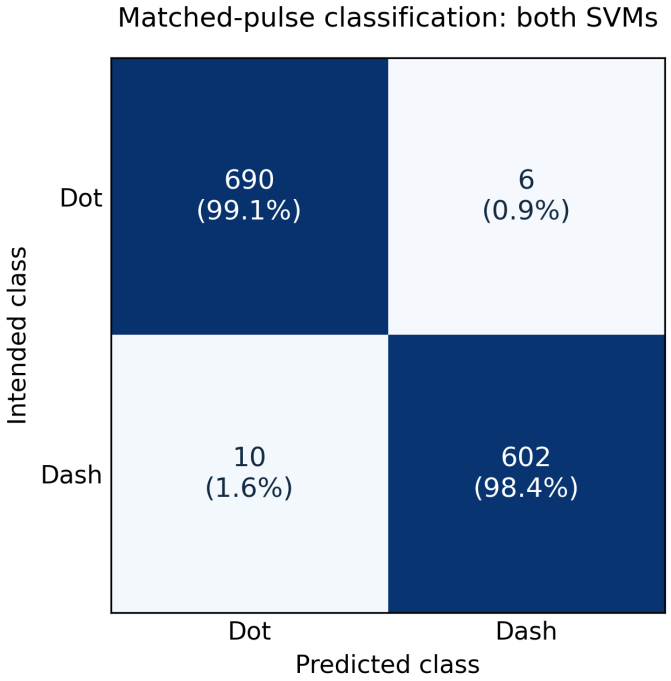


Figure 5.2: Dot/dash confusion matrix for the two SVMs on matched final-test pulses. Counts and row percentages are shown; each row sums to 100% before rounding. The two unmatched reference pulses are excluded from this conditional classification matrix.

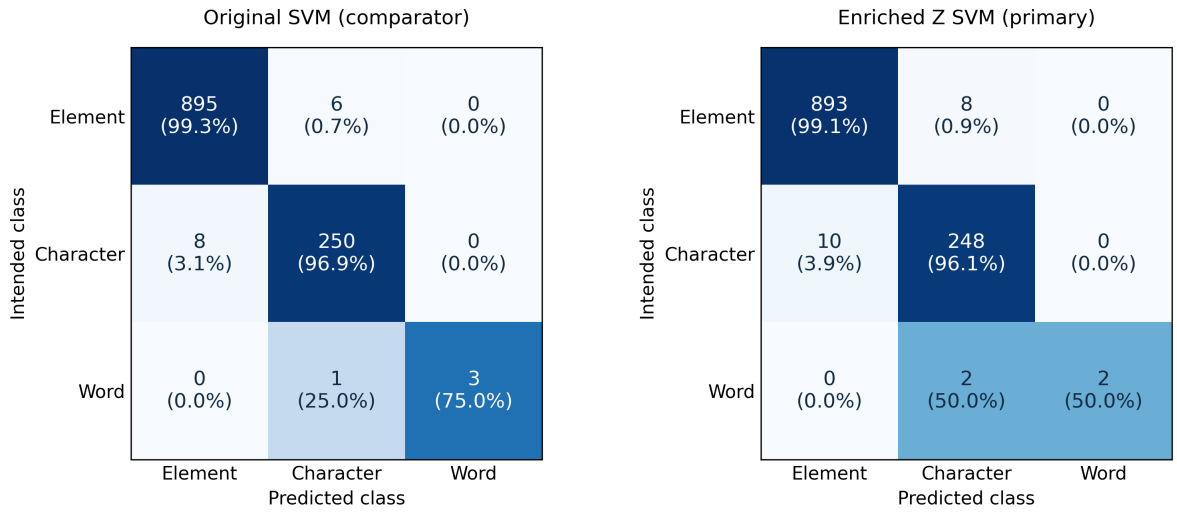


Figure 5.3: Gap confusion matrices for the original SVM comparator and enriched Z SVM primary candidate. Cells give counts and row percentages. Reference supports are 901 element gaps, 258 character gaps and only four word gaps. Gaps spanning unmatched reference pulses are excluded.

two more element-to-character, two more character-to-element and one more word-to-character decision.

Row percentages prevent the 901 element gaps from obscuring the less frequent classes. However, the original SVM's 75% and enriched SVM's 50% word-gap agreement are based on only four examples and should not be interpreted as stable estimates of word-boundary generalisation. All matrices compare predictions with intended/count-aligned labels; they do not independently distinguish operator encoding mistakes from decoder mistakes.

5.5 Performance by experimental condition

Table 5.5 gives the final CER breakdown for the original and enriched SVMs. Each group uses its own total reference-character denominator. The groups were not identical in target composition, and conditions were collected in a fixed block order.

Table 5.5: Descriptive final-test performance by session, instructed pace and command strength. Each strength contains 48 recordings. These grouped results do not isolate causal condition effects.

Grouping	Condition	Captures	Original SVM CER (%)	Enriched SVM CER (%)
Session	MAIN08	72	8.78	10.24
Session	MAIN09	72	9.27	12.68
Speed	slow	54	10.73	12.99
Speed	normal	45	5.79	4.96
Speed	fast	45	9.82	16.07
Strength	low	48	18.18	19.58
Strength	medium	48	3.01	4.51
Strength	high	48	5.22	9.70

Both SVMs had their highest strength-group CER at the low command. The original SVM's low-strength CER was 18.18%, compared with 3.01% at medium and 5.22% at high strength. The enriched SVM showed the same ordering. Pace-group CER did not vary monotonically: the normal condition had the lowest CER for both methods. These results report the recorded associations without attributing them exclusively to amplitude or pace.

5.5.1 Matched interference comparison

The full test contained 132 clean recordings and 12 table-tapping recordings. For the original SVM, these unbalanced groups had CERs of 8.42% and 14.29%, respectively. A more directly matched comparison used the 12 retained clean/noisy pairs, with 42 intended characters per condition (Table 5.6).

Table 5.6: Matched clean/table-tap comparison on the final test. Edit counts use the same 42 reference characters in each condition. Tap force was uncalibrated.

Method	Clean edits	Noisy edits	Clean exact	Noisy exact
Original RBF SVM	5/42	6/42	10/12	9/12
Enriched Z SVM (primary)	7/42	6/42	9/12	9/12

For the original SVM, two pairs were correct only in the clean condition and one only in the noisy condition. For the enriched SVM there were two in each direction. The small paired sample

showed no consistent large change across the two methods; it does not establish equivalence or immunity to interference.

5.6 Failed-message diagnostics

The original SVM failed to exactly recover 22 messages and the enriched SVM failed on 26. Table 5.7 partitions these failures into disjoint combinations of detection and intended-label disagreement. A single event disagreement can cause several text edits, so event counts and character edits are not interchangeable.

Table 5.7: Disjoint diagnostic categories among failed final-test messages. Labels indicate disagreement with intended/count-aligned references, not independently established decoder fault.

Diagnostic category	Original SVM	Enriched Z SVM
Detection only	2	2
Pulse label only	6	6
Gap label only	12	16
Pulse and gap labels	2	2
Total	22	26

Including mixed cases, gap-label disagreement appeared in 14 of the original SVM's failed messages and 18 of the enriched SVM's. Pulse-label disagreement appeared in eight failed messages for each. The original SVM produced eight invalid Morse tokens across eight messages; the enriched SVM produced nine across nine messages.

At the message level, enrichment reduced edit count on one capture, increased it on six and left it unchanged on 137. Four messages correctly recovered by the original SVM became incorrect with enrichment: three `SOS` captures and one `D` capture. No original-SVM failure became exactly correct. These observations locate the additional disagreements without establishing their physical or statistical cause.

Figure 5.4 illustrates three original-SVM failures. For intended `0`, all five pulses were detected but labelled as dots, producing `5`. Their reference holds were approximately 221–278 ms and intervening gaps 193–256 ms. For intended `M`, two detected dashes were separated by a predicted character gap, producing `TT`; the reference gap was approximately 635 ms. For `D6T1Z`, one unmatched reference pulse accompanied the prediction `N6T1Z`. The examples retain their original targets and scores.

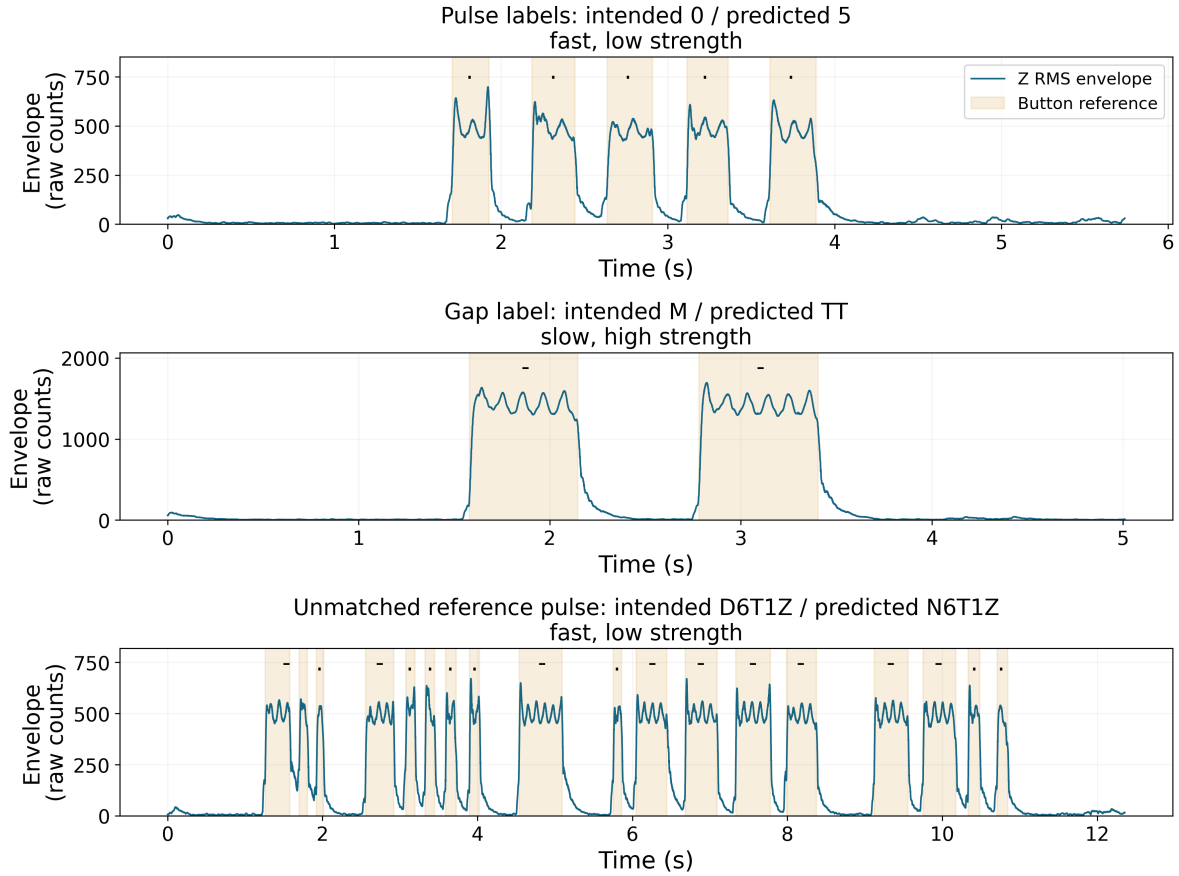


Figure 5.4: Representative final-test diagnostics for the original SVM. Shading denotes button-reference intervals; the symbols above pulses are model outputs. The examples illustrate pulse-label disagreement, gap-label disagreement and an unmatched reference pulse, respectively. They were selected for illustration, not to estimate prevalence.

5.7 Acceptance evidence and outstanding measurements

Table 5.8 summarises the available evidence against the ATP identifiers. "Supported" means the stated investigative evidence was produced; it does not imply that an undefined numerical target or a broader original scope commitment was satisfied. The numerical targets and scope amendments must be interpreted consistently with Chapter 2.

Table 5.8: Acceptance-evidence summary for the completed offline investigation. Unverified items are retained explicitly.

ATP	Evidence status	Recorded outcome or limitation
ATP1	Supported, bounded	Acquisition audit and event-matching results; no proven alias-free or minimum-rate claim.
ATP2	Supported	Pulse/gap confusion matrices with matched-event denominators and label limitations.
ATP3	Supported, target unresolved	Text outputs, CER and exact recovery; original 90% "accuracy" benchmark requires a defined metric.
ATP4	Supported	IMU/timestamp prediction inputs, without microphone, key or target features.
ATP5	Supported within scope	Single-operator instructed-speed/session results; no between-user or validated 30% variability envelope.
ATP6	Supported, bounded	Twelve final-test matched clean/table-tap pairs; uncalibrated interference.
ATP7	Supported	Raw attempts retained; manifest, inclusion decisions and 638 primary captures documented.
ATP8	Supported	Fixed session-separated partitions, finite representations and training-only scaler/fit checks.
ATP9	Supported for revised scope	Duration baseline and LR/SVM/RF compared; CNN/recurrent approaches not implemented.
ATP10	Supported	Frozen methods and scoring evaluated on 144 reserved captures without retuning.
ATP11	Partial	Saved artifacts and earlier host timings available; consistent resource benchmark and peak RAM not established.
ATP12	Unverified	No causal acquisition-to-displayed-text latency test or demonstrated 1 s/character target.
ATP13	Supported computationally	Code/configuration/checksum records and local reproduction; physical remounting repeatability unverified.

No final-test runtime benchmark, embedded implementation, isolated actuator rise-time measurement, controlled sampling-rate reduction study or physical remounting experiment was completed. The results therefore establish performance of the recorded, bounded offline decoding task under the stated reference and cohort limitations. Broader interpretation and recommendations are developed in the Discussion chapter.